

Movement Augmentation in Virtual Reality: Impact on Sense of Agency Measured by Subjective Responses and Electroencephalography

Liu Wang^{1,3}, Mengjie Huang^{1*}, Chengxuan Qin², Yiqi Wang⁴, Rui Yang²

1) Design school, Xi'an Jiaotong-Liverpool University 2) School of Advanced Technology, Xi'an Jiaotong-Liverpool University

3) Department of Civil Engineering and Industrial Design, University of Liverpool 4) UCL Interaction Center, University College London

ABSTRACT

Virtual movement augmentation, which refers to the visual amplification of remapped movement, shows potential to be applied in motion-related virtual reality programs. Sense of agency (SoA), which measures the user's feeling of control in their action, has not been fully investigated for augmented movement. This study investigated the effect of augmented movement at three different levels (baseline, medium, and high) on users' SoA using both subjective responses and electroencephalography (EEG). Results show that SoA can be boosted slightly at medium augmentation level but drops at high level. The augmented virtual movement only helps to enhance SoA to a certain extent.

Keywords: Sense of Agency, Movement Augmentation, EEG

Index Terms: Human-centered computing—Human computer interaction (HCI) —Interaction paradigms—Virtual reality

1 INTRODUCTION

When designing movement representation in virtual reality (VR), users' virtual movements are often expected to map with their real ones in the physical world. For VR applications targeted for specific purposes (such as exercise and rehabilitation), movement augmentation can be applied [1]. Virtual movement augmentation can be realized with strategies such as the spatial manipulation of position in a virtual hand or rotation angle of a virtual arm. As there is a mismatch between the user's real motion and virtual motion in virtual movement augmentation, the users' experience of control may vary. Sense of agency (SoA), which refers to the experience of controlling one's action and the external environment, is a critical consideration in human-computer interface (HCI) [2]. Studying the users' SoA with movement augmentation is essential for ensuring a fundamental experience of conscious movement in VR. In addition, investigating how SoA can be boosted through augmented movement is meaningful to VR rehabilitation applications, where improved SoA is preferred from the rehabilitation perspective.

In VR, the SoA has not been fully investigated for motion-related tasks, especially for augmented movement [3], [4]. Prior studies investigated mainly SoA in a single action-effect paradigm in non-VR condition, and the subjective responses of the controlled condition is a frequently used method [1]. Empirical evidence remained unclear on whether the augmented movement can help enhance SoA [3], [4]. Previous studies have shown that the neurophysiological signals can reflect people's SoA. For example, stronger SoA of voluntary hand movement is indicated by a stronger power decrease of the alpha band in central and parietal areas when compared with lower SoA of movement performed that are not voluntary [5].

This research aims to investigate users' SoA with augmented movements in VR by using both subjective responses and the neurophysiological signals of EEG in a controlled experiment. The study results can contribute to design guidelines for motion-based VR programs with movement augmentation, such as virtual rehabilitation and exergames.

2 METHOD

The experiment was designed with three augmentation modes for reaching movement in VR. We adopt the self-report of SoA and the measure of brain activities that can reflect SoA in this experiment for analyzing both subjective responses and objective neurophysiological cues of augmented movement.

2.1 Virtual Movement Augmentation

Figure 1 illustrates movement augmentation in virtual reaching. Movement augmentation is achieved by manipulating the virtual hand reaching position in real-time. The forward distance of the virtual hand was visually longer than the reaching distance of the real hand. The level of augmentation can be adjusted by a scale factor S , which is determined as follows:

$$S = \frac{P_v}{P_r} \quad (1)$$

P_v refers to the virtual hand's moving distance from the starting line and P_r refers to the real hand's moving distance from the starting line. Three augmentation modes were implemented in VR reaching movement in this study: (1) *Baseline*: $S = 1$, (2) *Medium augmentation*: $S = 1.5$, and (3) *High augmentation*: $S = 2$.

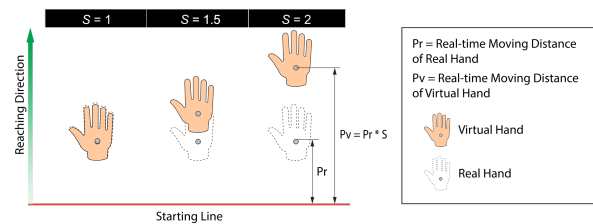


Figure 1: Movement augmentation in virtual reaching

2.2 Experiment

Ten subjects (8 male, 2 female) aged between 19-27 (mean = 23.6, SD = 4.3) were recruited in the experiment. All of them are healthy without neurological and motor issues, and they all signed a written consent form before the experiment. The subjects had to finish three reaching tasks arranged in random order. Each task was implemented with an augmentation mode, and needed the subjects to reach 15 times. As shown in Figure 2 (A), the reaching tasks in the virtual environment (developed by Unity engine) required the subjects to reach his/her right hand to push the box on the desk to collide the wall 30 centimeters away from the starting line.

* Corresponding author: mengjie.huang@xjtlu.edu.cn

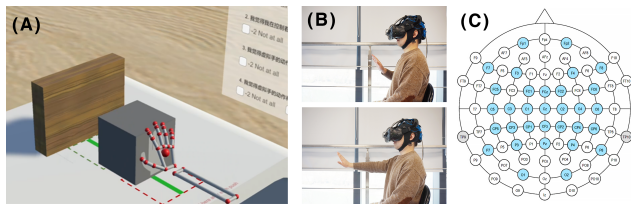


Figure 2: Virtual environment (A); reaching task for subjects wearing HMD and EEG cap (B); and the EEG channel localization (C)

2.3 Measure of SoA

2.3.1 Self-rating Scales of SoA

After each task, the subjects were required to rate their experienced SoA of the virtual reaching movement using a 5-point Likert scale from totally disagree (scored -2) to totally agree (scored 2). The scoring of questions regarding SoA in each task were calculated as a sum [6]. The four questions are: (Q1) *The movements of the virtual hand felt like they were my movements.* (Q2) *It felt like I was controlling the movements of the virtual hand.* (Q3) *It felt like I was causing the movements of the virtual hand.* (Q4) *The movements of the virtual hand were in sync with my own movements.*

2.3.2 EEG Measure of Agency

All subjects' EEG data were recorded during the reaching tasks. As shown in Figure 2 (B), the EEG cap from Emotiv Flex (Saline Kit) with 32-channel was used while they were also wearing the VR head-mounted display (HMD). The data were recorded at a sampling rate of 128 Hz. The locations of the channels were based on an international 10-20 EEG location system and the selected channels are shown in Figure 2 (C). The EEG data were preprocessed offline using MATLAB and functions provided in EEGLAB to avoid noise in the data. The EEG data were filtered with bandpass of 1-50 Hz, and were processed using independent component analysis to remove the artificial components. After the preprocessing, the data in each task were analyzed using Hilbert-Huang transform to calculate the average power in the alpha band.

2.4 Self-ratings

The results of self-ratings of SoA are shown in Table 1. The highest rating of SoA in mean value is found in the *medium augmentation* mode ($M = 6.90$, $SD = 1.85$), followed by *baseline mode* ($M = 6.70$, $SD = 1.70$). The lowest rating of SoA is found in the *high augmentation* mode ($M = 5.00$, $SD = 2.87$).

In the high augmentation mode, the subjects did not feel the strongest SoA; rather they experienced a weaker SoA in controlling their virtual hand as compared with the other two modes. The augmentation of virtual hand reaching doesn't necessarily help to boost the SoA all the time, which is an unexpected phenomenon. According to One-Way ANOVA, no significant differences in rating results were found in the three modes ($p = 0.125$).

Table 1: Self-ratings of sense of agency

Augmentation modes	S=1	S=1.5	S=2	p
Mean	6.70	6.90	5.00	0.125
SD	1.70	1.85	2.87	

2.5 Neurophysiological analysis of agency

The results from the analysis of 10 subjects (as shown in Figure 3) showed that the normalized average power in the alpha band was the lowest in CPz ($M = 0.25$, $SD = 0.26$) and Pz ($M = 0.40$, $SD =$

0.28) electrodes in $S=1.5$ mode. One-Way ANOVA showed no significant differences among the impact of augmentation factors on the average power (CPz: $p > 0.01$; Pz: $p > 0.01$).

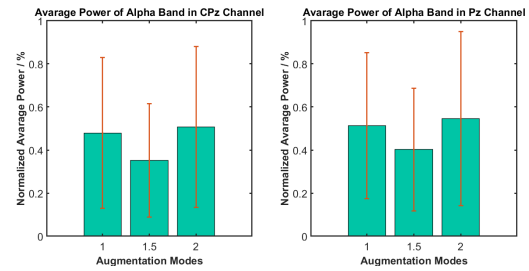


Figure 3: Normalized average power of alpha band of different movement augmentation modes: in CPz (left) and Pz (right) channels

Though not significant, the higher SoA can be reflected in brain activities in certain electrodes in the central-parietal area by the lower average power in the alpha band. The results are in line with some of the previous research work that investigated SoA of hand movement in non-VR conditions [5]. The visually seen augmented movements may have altered subjects' sensorimotor processing, as the results of alpha power showed sensitive to the changes in SoA.

3 CONCLUSION

This study investigated how augmented movement in VR influenced human's experience of agency by using both self-report ratings and EEG. The results showed that augmented movement at a certain level could induce slightly higher SoA. However, the increase in augmentation of movement doesn't necessarily help to produce higher SoA when movements are visually augmented excessively. The EEG analysis showed that SoA of augmented movements are linked with the changes of alpha rhythm in parietal cortex, which are associated with sensorimotor processing.

4 ACKNOWLEDGMENT

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